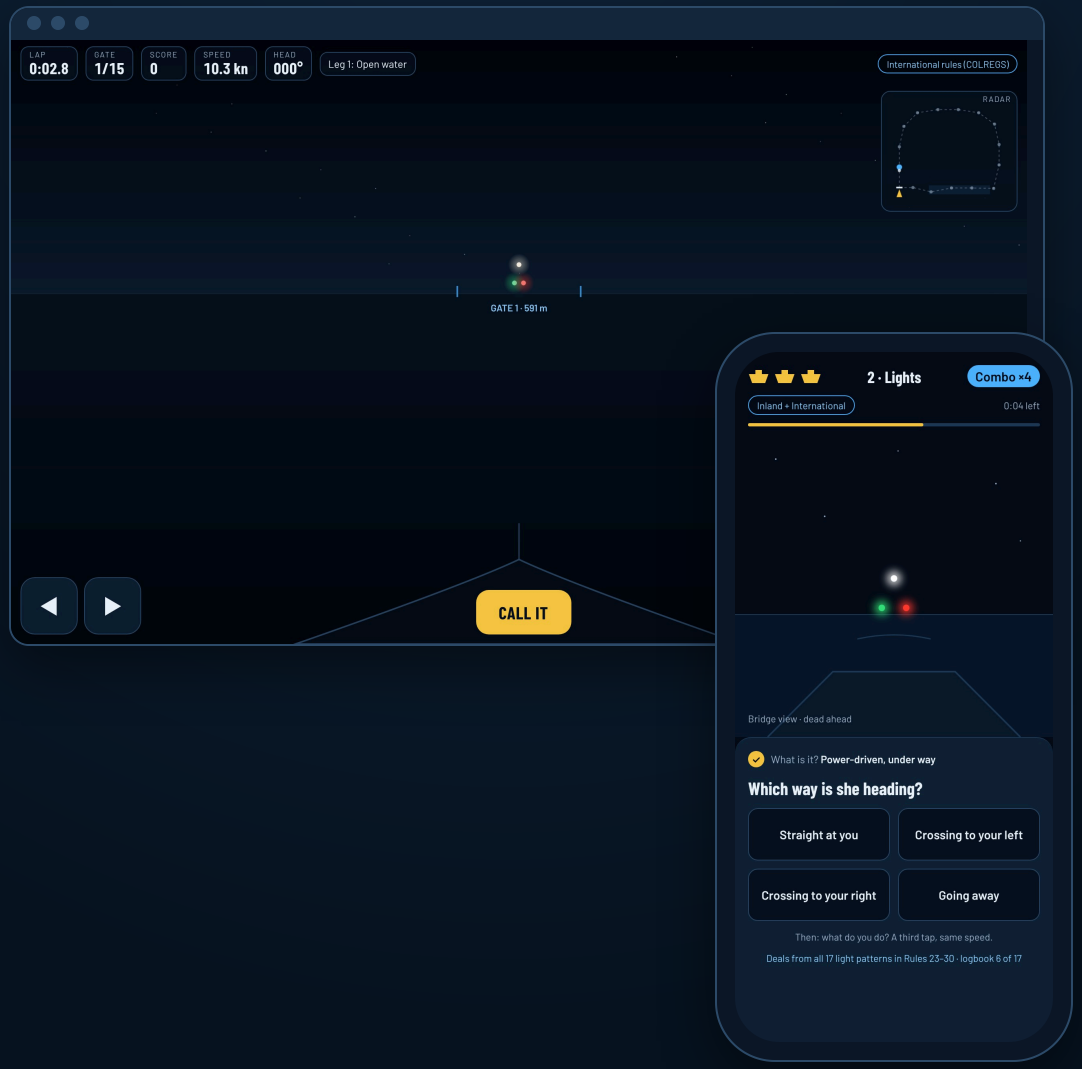
CAPTAIN'S ACADEMY · ARCADE GAME

Rules of the Road prototype book

Every prototype from the design and build sessions of 21 to 25 October 2026, cleaned up and in build order: Chapter 1 screen mockups, the content decks, the Harbor Run course, the playable Night Harbor Run, and the first two levels in the real game.

Mockup a picture of the idea **Prototype** playable, outside the repo **Real build** in the repo, with tests



CONTENTS

What is in this book

The order follows how the work was made: plan, screens, content, course, playable prototype, then the real game.

1 The chapter map Mockup

2 Rapid-fire levels 1 to 5 Mockup

3 Feedback and the boss Mockup

4 Steering levels 6 to 8 Mockup

5 Lights deck Mockup

6 Day shapes deck Mockup

7 Harbor Run course Mockup

8 Night Harbor Run Prototype

9 Level 3 harbor course Real build

10 Chapter 1, level 1 Definitions Real build

11 Status and next steps

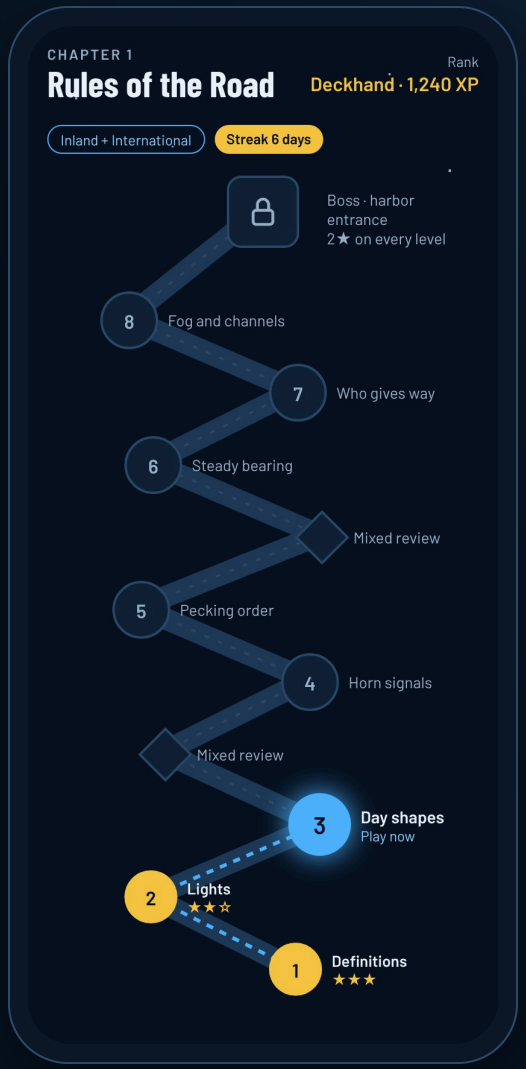
Cleaned up for this book

- Every screen re-rendered at double resolution and framed as the device it plays on.
- Definitions mockup: the progress counter now reads 7 of 17 cards, matching the deck.
- Night Harbor Run: the gate distance label moved below the waterline so it no longer covers vessel lights; the leg name sits in a readable pill on the daylight leg.
- Night Harbor Run: a link ending in `#gate10` (any gate 1 to 15) now starts the lap at that gate, which is how the day, channel and fog legs were captured.
- Mockups are proposals. Rule citations and sample questions still need the uscg-regs check.

1 · THE CHAPTER MAP Mockup

Chapter 1 is a channel into port

Eight levels and a boss, in the order DECISIONS.md G7 sets, with mixed-review stops between them.



Chapter map

Cleared levels gold, next level glowing, boss locked at the harbor entrance.

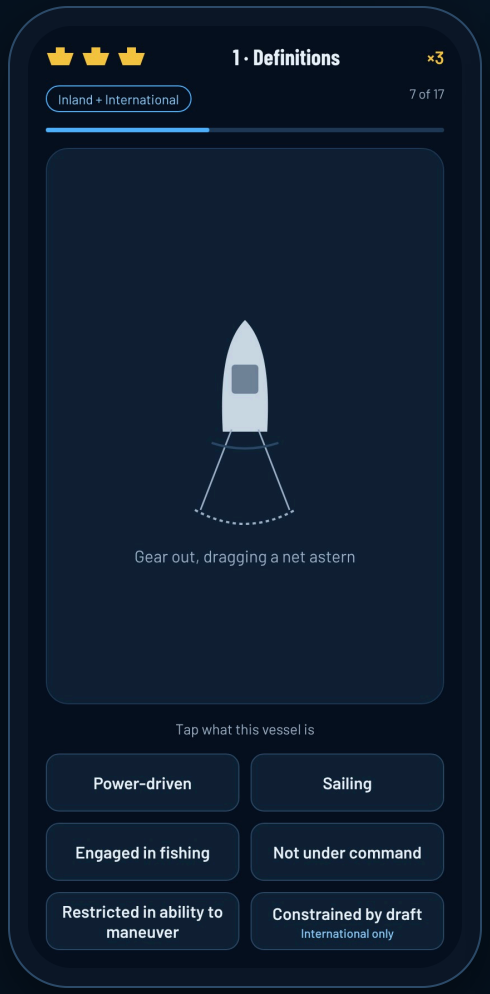
The levels

- **1 Definitions** · Rule 3 · upright
- **2 Lights** · Rules 21 to 31 · upright
- **3 Day shapes** · Rules 20 to 30 · upright
- **4 Horn signals** · Rules 32 to 35 · upright
- **5 Pecking order** · Rule 18 · upright
- **6 Steady bearing** · Rule 7 · sideways
- **7 Who gives way** · Rules 13 to 17 · sideways
- **8 Fog and narrow channels** · Rules 9, 10, 19 · sideways
- **Boss** · exam format, 90% on two runs (G10)
- Upright levels play with one thumb; sideways levels put a lever under each thumb (G8).

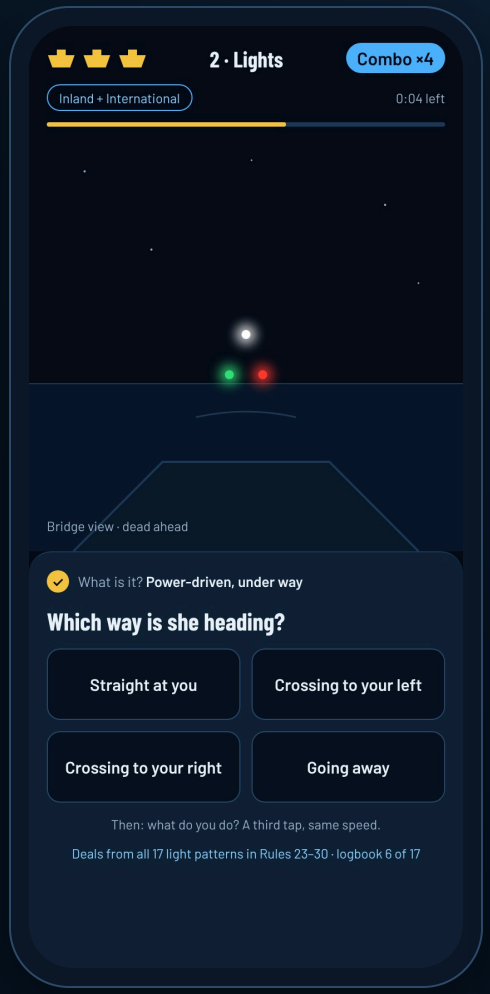
2 · RAPID-FIRE LEVELS Mockup

Levels 1 to 4: name it fast, one thumb

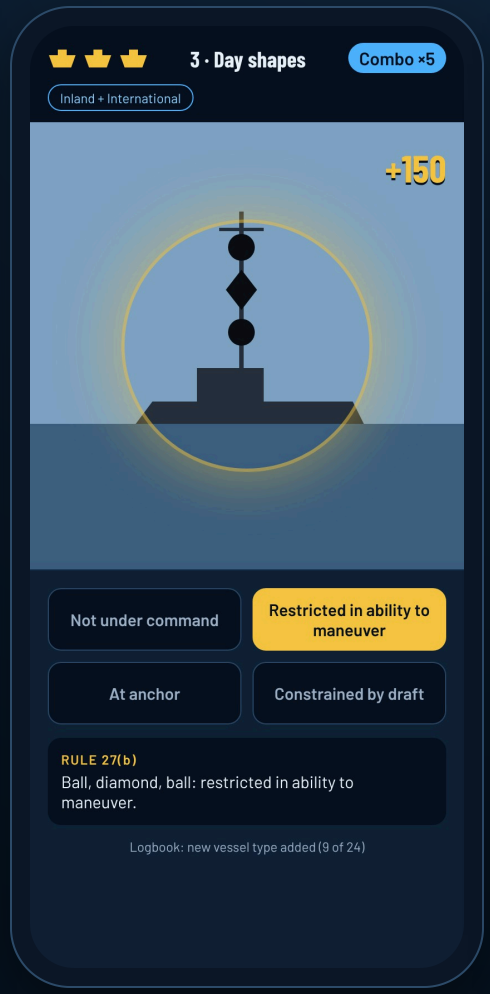
Many quick visual examples with instant feedback, until recognition is automatic (G20).



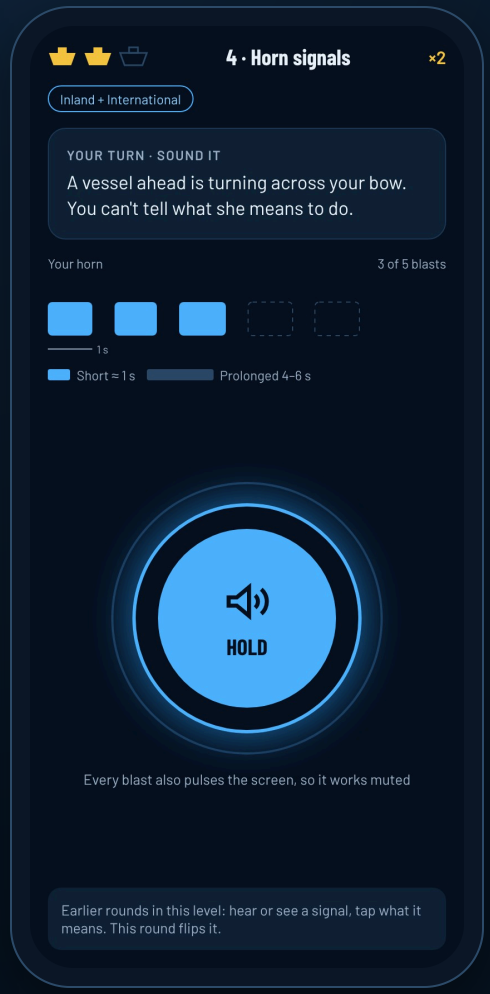
1 · Definitions
Tap what the vessel is, from six Rule 3 terms.



2 · Lights
What is she, then which way is she heading.



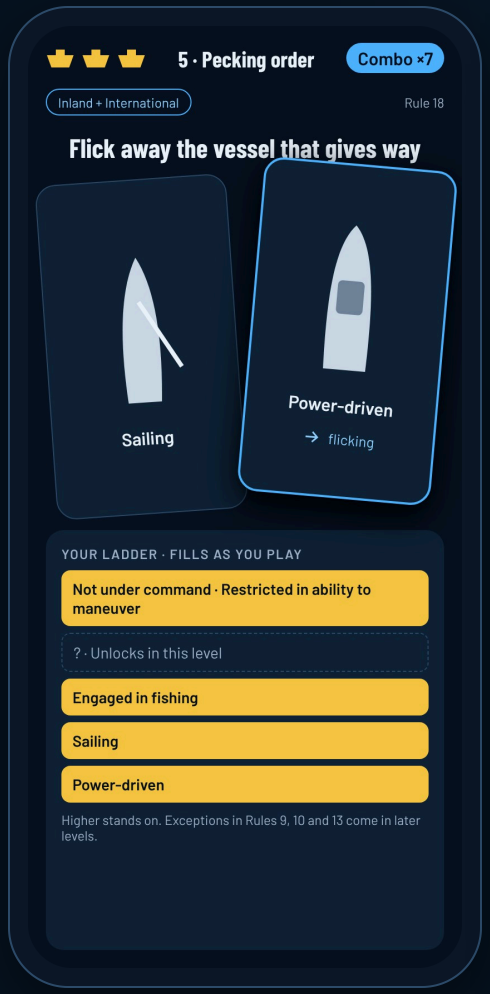
3 · Day shapes
A right answer: gold flash, points and the rule line.



4 · Horn signals
Hold to sound five short blasts at real lengths.

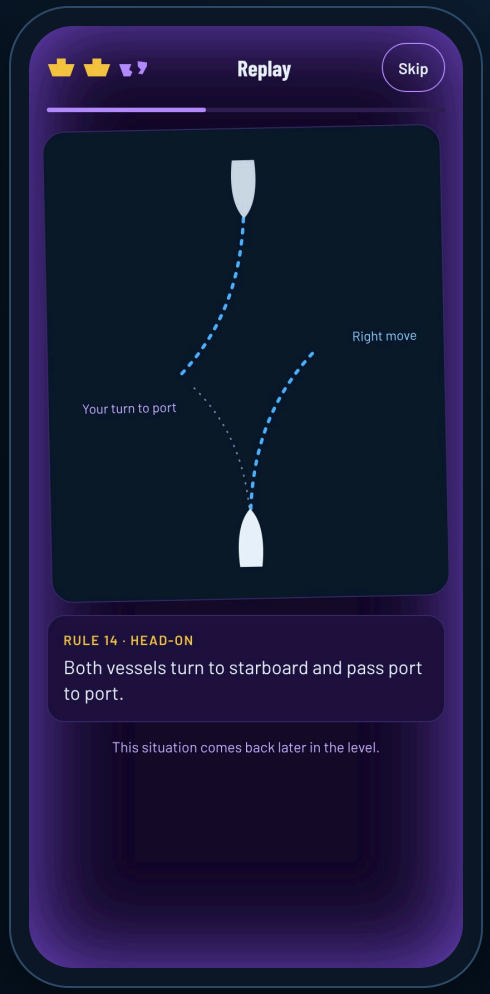
Level 5, a miss, and the chapter boss

A miss costs a hull heart and plays a replay of about five seconds that shows the right move and cites the rule (G11).



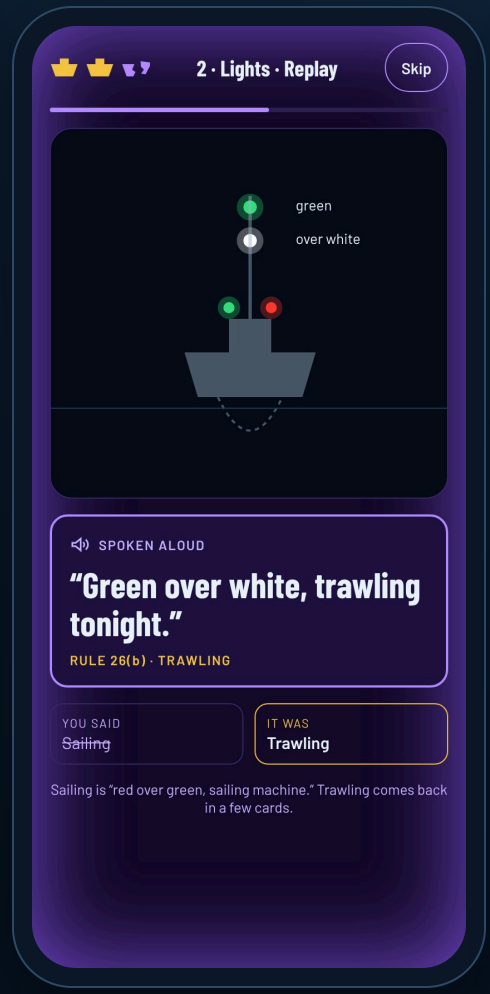
5 · Pecking order

Flick away the vessel that gives way; the Rule 18 ladder fills as you play.



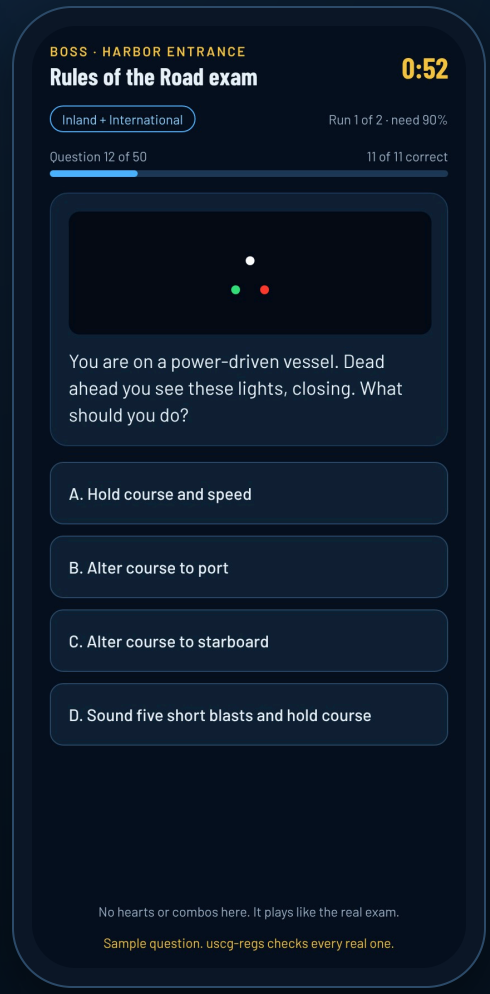
A steering miss

Head-on replay: both turn to starboard, Rule 14.



A lights miss

The pattern is said aloud: "green over white, trawling tonight."

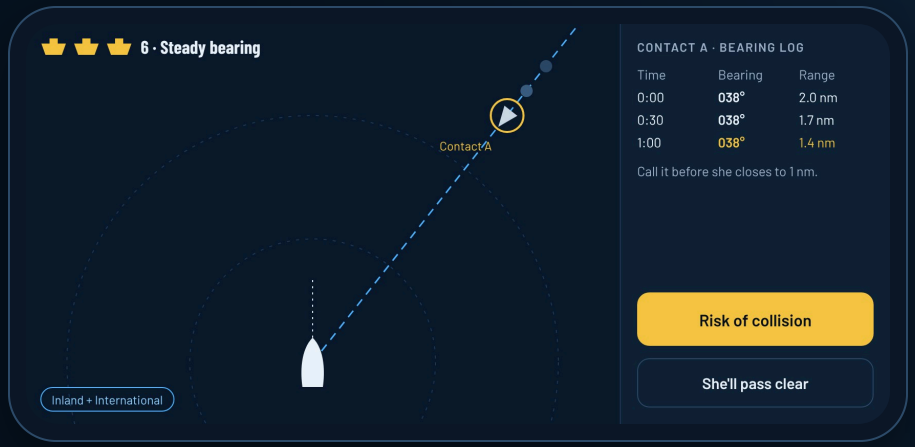


The boss

Timed multiple choice in exam format, 90% to clear.

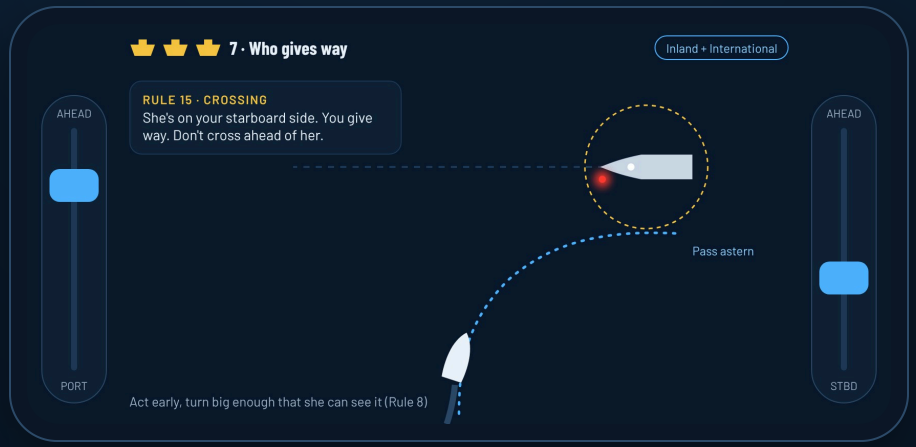
Levels 6 to 8: apply the rule at the helm

Played sideways with the Long Beach prototype's twin engine levers.



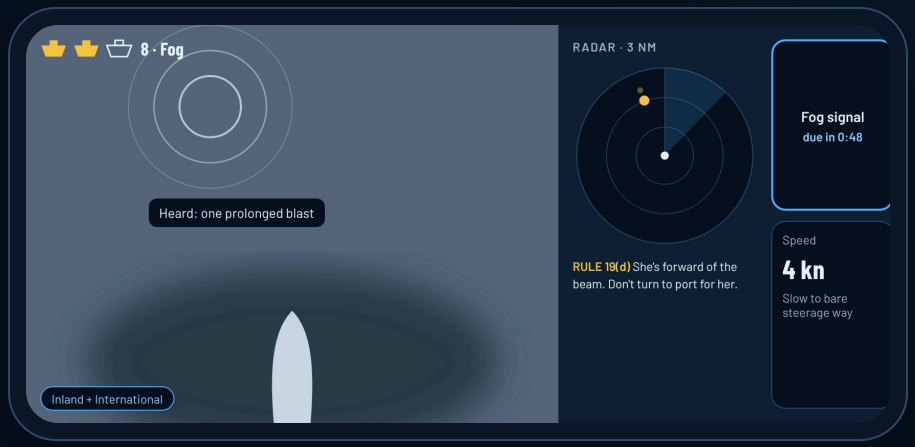
6 · Steady bearing

Same bearing, closing range: call the risk of collision (Rule 7).



7 · Who gives way

She is on your starboard side: turn early and pass astern (Rule 15).



8 · Fog

Radar and a horn pulse only; slow down, no turn to port (Rule 19).

Shared on every screen

- Three hull hearts and the rule-set chip (G6).
- Interface in gold, blue and purple only; red and green are navigation lights (G15).
- Every horn also shows as a visual pulse (G16).

 Mockup

Every vessel's lights in Rules 23 to 30

The 17 patterns the Lights level deals from, each seen from ahead, abeam and astern.



How it plays

- Only lights whose Rule 21 arcs face the player are drawn: masthead 225°, sidelights 112.5° each, sternlight 135°, all-round 360°.
- A miss is spoken in a short phrase and shown in words.
- Inland-only patterns (pushing ahead, special flashing light) go to a later Differences level.
- Draft list: uscg-regs checks every pattern.

 Mockup

Every ball, cone, diamond and cylinder

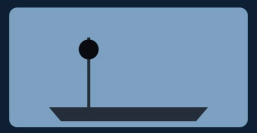
By day, shapes replace lights. Each one becomes a vessel to read at speed.

DAY RUN · THE SHAPES DECK

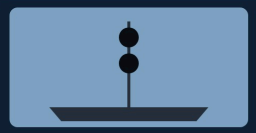
Every ball, cone, diamond and cylinder

By day, these shapes replace the lights. Each one is a boat parked or moving on the Harbor Run course, and the player has to read it at speed and do the right thing.

Draft list. uscg-regs checks every shape and citation against the rule text before it's built.



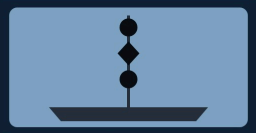
At anchor
One ball, forward
Rule 30(a)



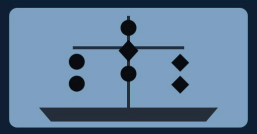
Not under command
Two balls. She can't get out of your way
Rule 27(a)



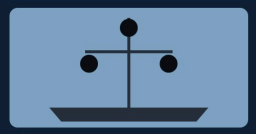
Aground
Three balls. Shoal water nearby
Rule 30(d)



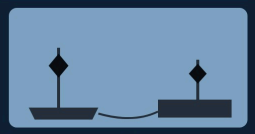
Restricted in ability to maneuver
Ball, diamond, ball
Rule 27(b)



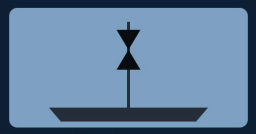
Dredging or underwater work
Balls on the blocked side, diamonds on the side you pass
Rule 27(d) · "Pass on the diamonds"



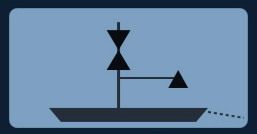
Mine clearance
Three balls in a triangle. Stay 1,000 m clear
Rule 27(f)



Towing, tow over 200 m
A diamond on the tug and on the tow. Never cut between them
Rule 24(a)



Engaged in fishing
Two cones, points together. The hourglass
Rule 26



Fishing, gear out over 150 m
Add a cone point up, on the side the gear runs
Rule 26(c)



Motor-sailing
Cone point down. Sails up, engine on: treat her as power-driven
Rule 25(e)



Constrained by draft
One cylinder. Don't impede her
Rule 28 · International only



Small vessel, divers down
A rigid copy of flag A in place of the shapes
Rule 27(e)

Worth remembering

- One ball at anchor, two not under command, three aground.
- Ball, diamond, ball: restricted in ability to maneuver.
- Dredge: pass on the diamonds, never the balls.
- Cone point down: motor-sailing, so treat her as power-driven.
- Cylinder: constrained by draft, International only.

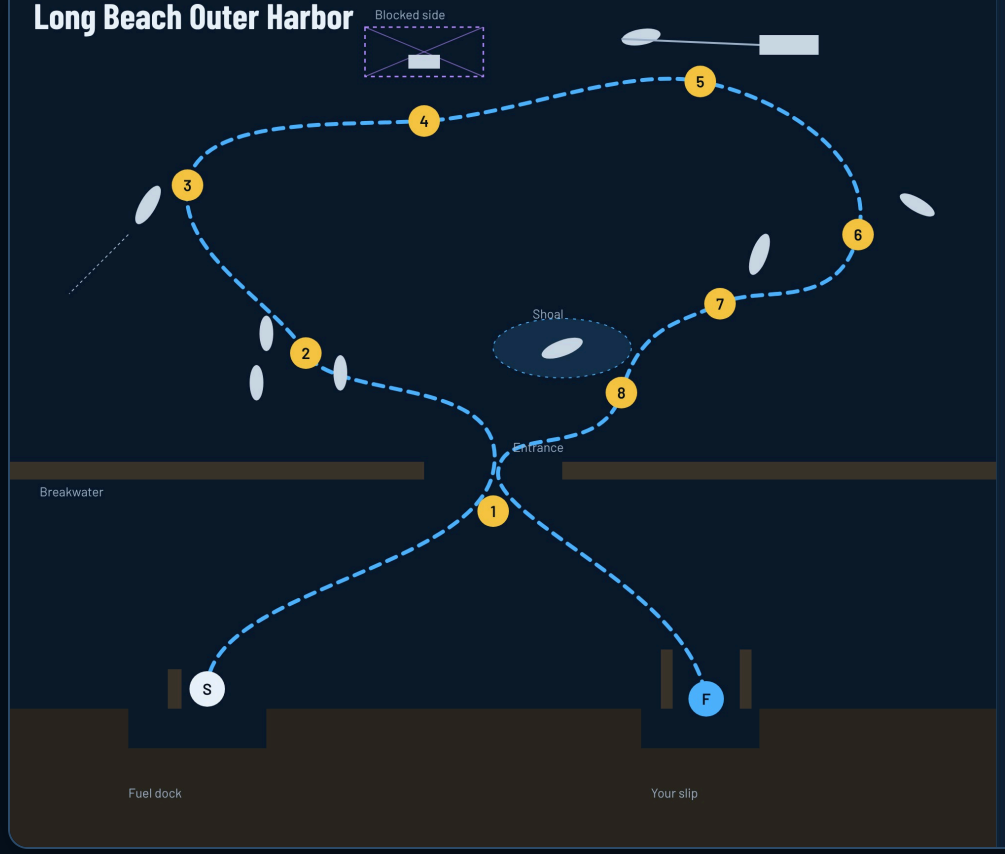
7 · HARBOR RUN COURSE **Mockup**

A race course on the Long Beach prototype

Eight checkpoints, each a vessel to read and a rule to act on, then dock against the clock.

CHAPTER 1 · HARBOR RUN

Long Beach Outer Harbor



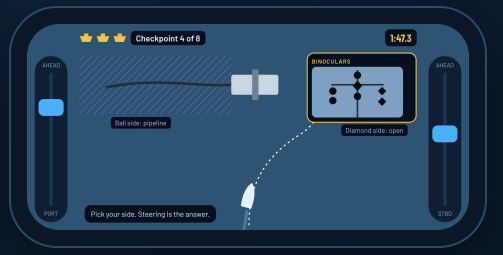
Day run · shapes | Night run · lights

Par 3:30 · 8 checkpoints — Best: 3:12 ★★★

1. **Exit the breakwater.** Keep to the starboard side of the entrance. Rule 9
2. **Anchored fleet, one ball.** Weave through and call "at anchor". Rule 30
3. **Hourglass trawler.** Give way and keep off her gear. Rules 18, 26
4. **Dredge, the fork.** Pick the diamond side. The ball side is blocked. Rule 27(d)
5. **Tug and tow, diamonds.** Pass astern of the tow, never between. Rule 24
6. **Motor-sailer, cone down.** The trick: she's power-driven, crossing from starboard. You give way. Rules 15, 25(e)
7. **Two balls, drifting.** Not under command. Give her room. Rule 27(a)
8. **Three balls, aground.** The shoal she's on is your hazard too. Rule 30(d)

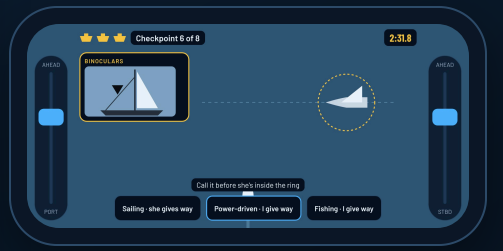
F **Back in and dock.** The first prototype's slip, under the touch-speed limit

A wrong call or a wrong move costs a hull heart and plays the 5-second replay; the clock keeps running. Hitting a boat or cutting a tow ends the run. Each run reshuffles which vessel sits at which checkpoint.



Pass the dredge

Steering is the answer: the diamond side.



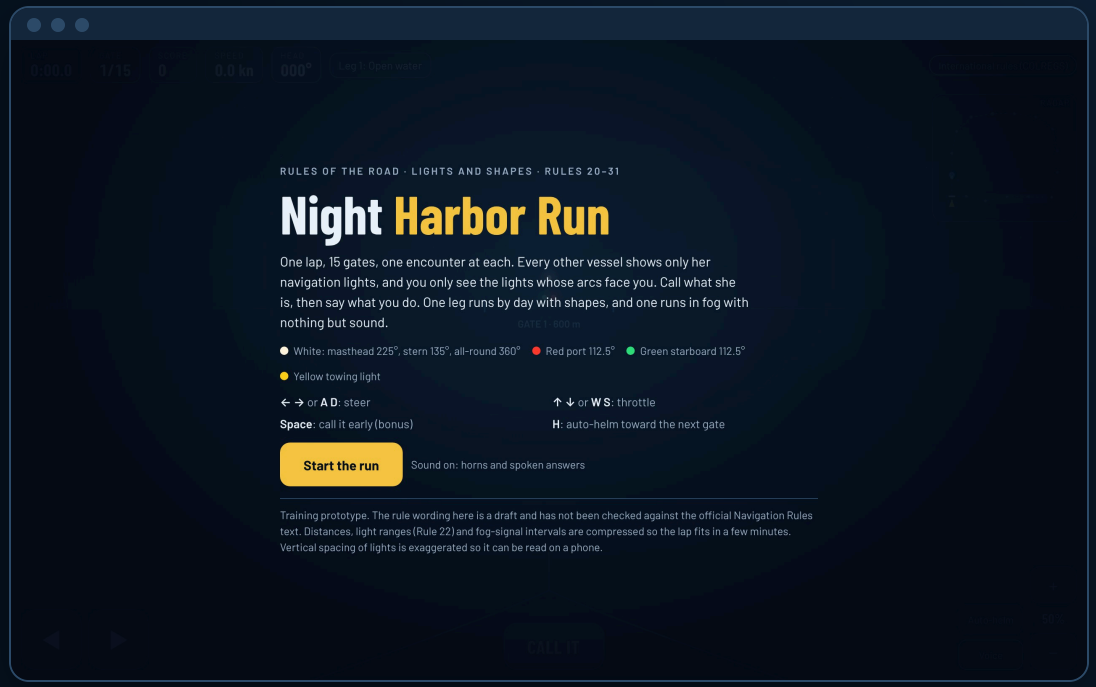
Call it while steering

Motor-sailer: she is power-driven.

8 · NIGHT HARBOR RUN **Prototype**

One lap at night, 15 gates, lights only

A single HTML file that runs on desktop and phone with no backend. The playable version of the course idea.



Start screen

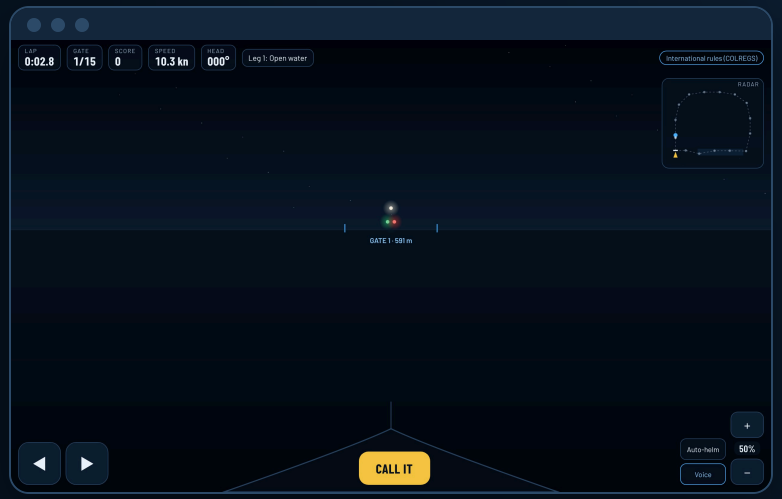
Light colors and arcs, controls, and the draft note.

How it works

- At each gate a vessel shows only the lights whose arcs face you, computed from her heading and your bearing.
- The game pauses near her, or early with CALL IT for a bonus: what is she, what do you do, which signal.
- Feedback gives the right answer, the rule and a spoken phrase.
- One leg runs by day with shapes; one runs in fog by sound alone.
- International rules throughout. Distances and fog intervals are compressed.

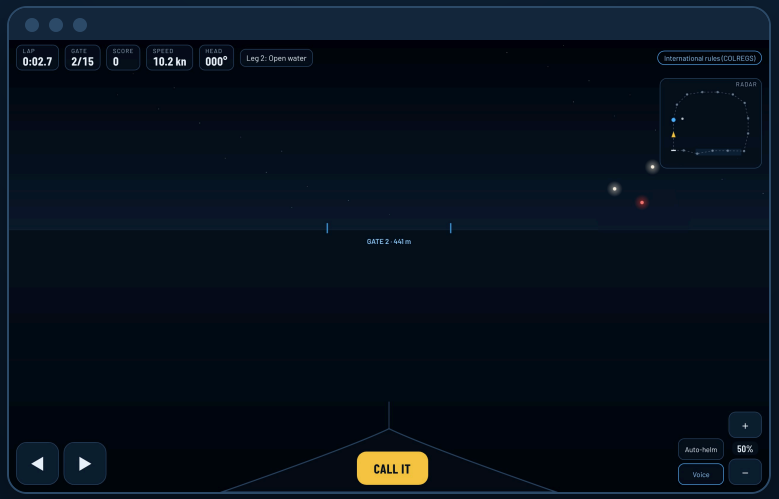
What you see at the gates

The lights change correctly as you move around a vessel.



Head-on

Masthead over red and green: alter to starboard, one short blast.



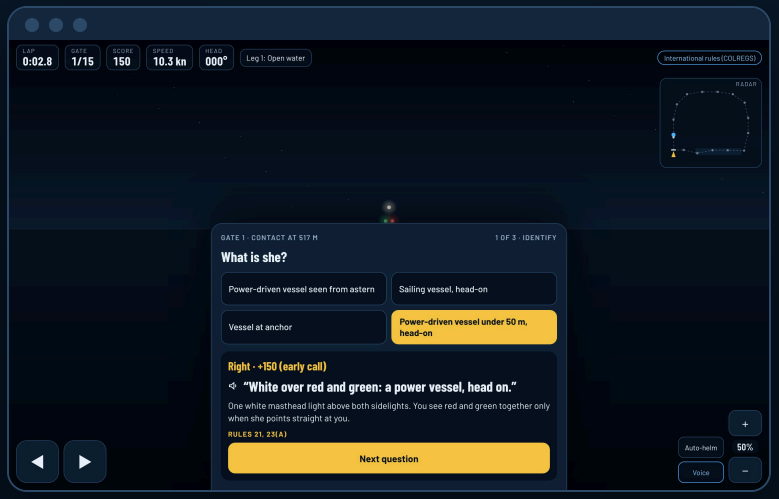
Crossing from starboard

Two masthead lights and red: a ship 50 m or more, you give way.



Sailing vessel

Red over green at the masthead, no masthead light: power gives way.

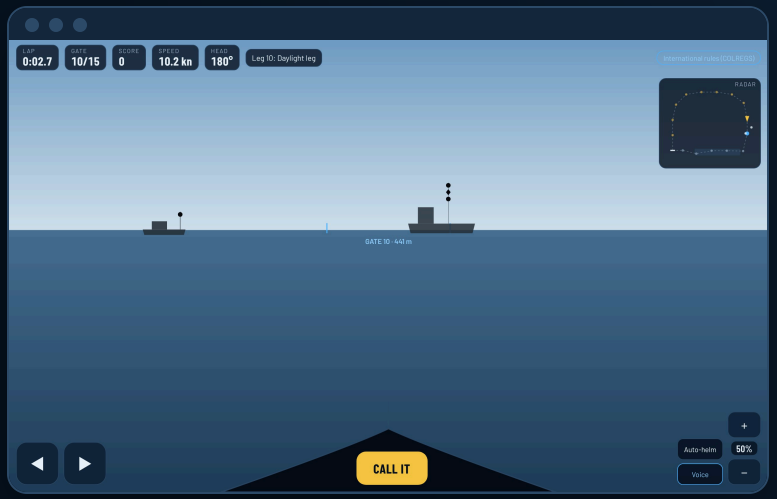


An answer

Right, with the early-call bonus, the spoken phrase and Rules 21, 23(a).

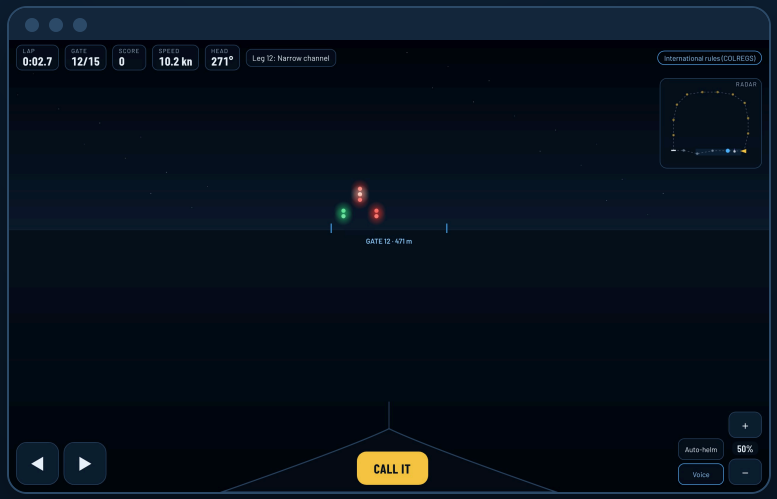
The special legs

Captured by opening the file with #gate10, #gate12, #gate14 and #gate15.



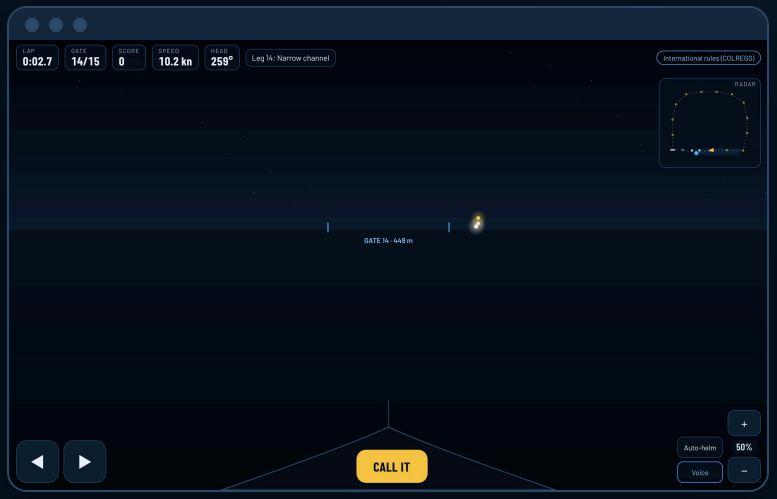
Day leg, gate 10

Ball, diamond, ball on a buoy tender; one ball on a boat at anchor.



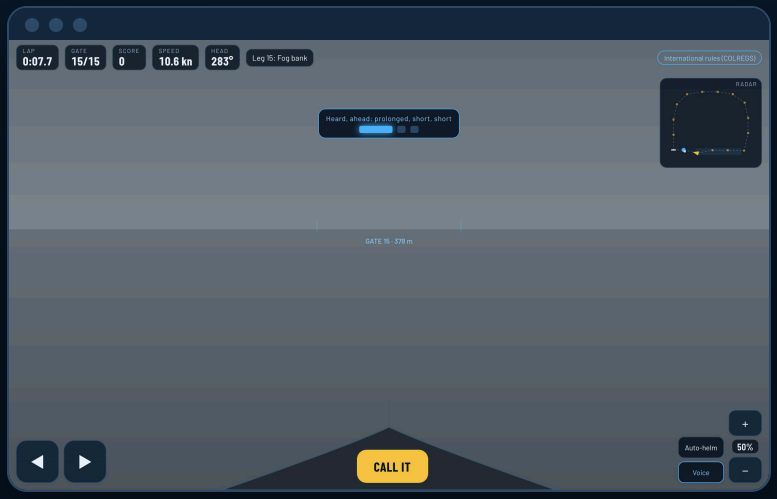
Narrow channel, gate 12

Dredge: red-white-red, two reds blocked, two greens passable.



Tug and tow from astern, gate 14

Yellow towing light over a sternlight, and the tow's sternlight nearer.

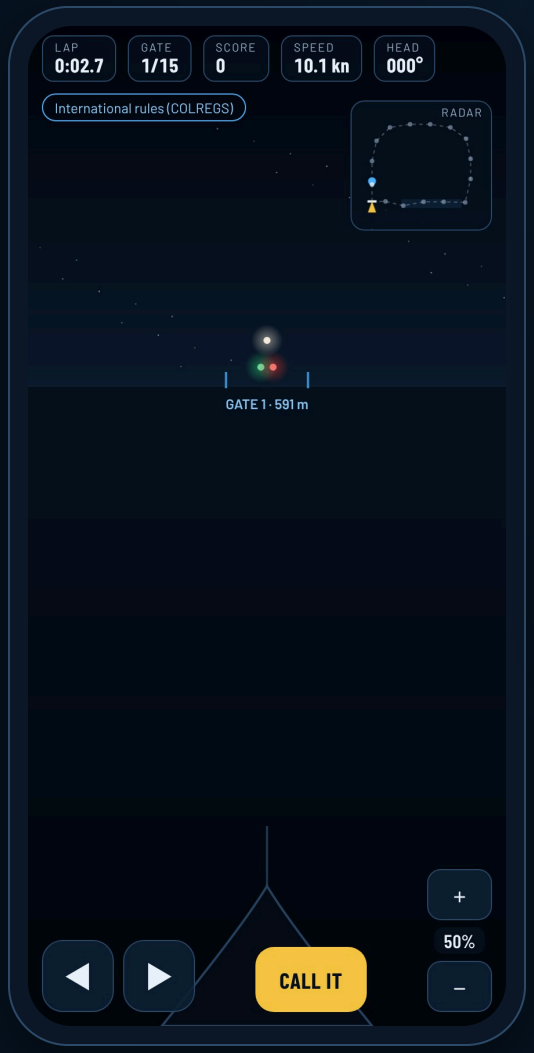


Fog leg, gate 15

Nothing visible; one prolonged, two short, heard ahead and shown as a pulse.

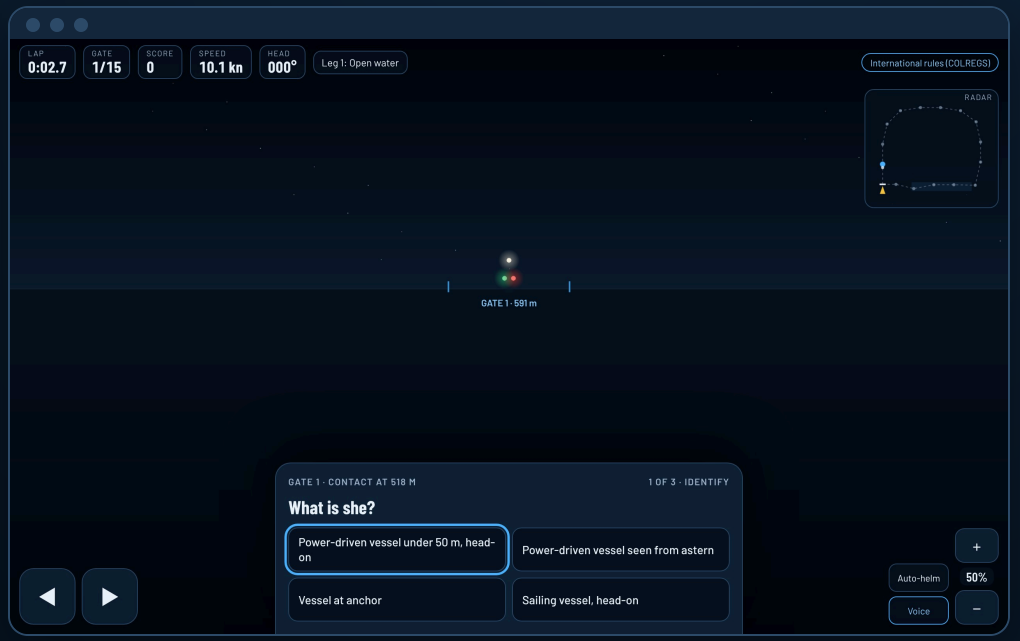
On a phone, and the encounter questions

The same file adapts to a phone held upright.



Phone, upright

Bridge view, radar, steering, throttle and CALL IT.



Called early

Step 1 of 3: identify. Then the action, then the sound signal.

 Real build

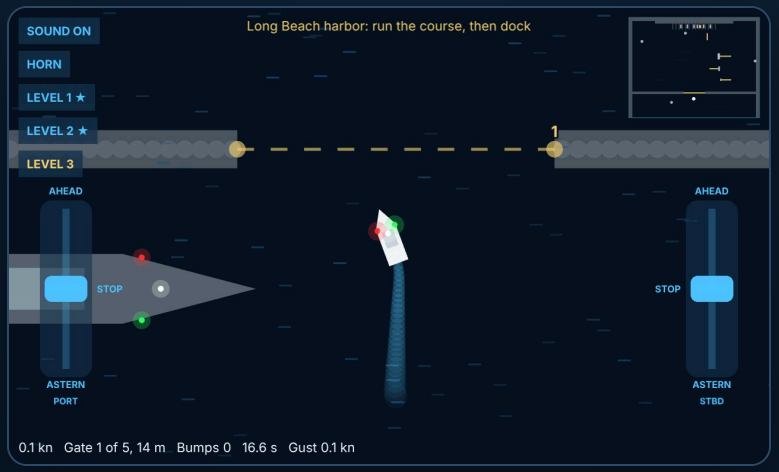
Gates, moving traffic, and a wrong turn gets you hit

Branch head-dev/harbor-traffic-course, issue #46. Screens from the production build.



The start

Gate 1 is the breakwater gap; the HUD counts gates.



The container ship

A 60 m ship crosses outside the breakwater from the start.



Hit by the ship

The banner names the vessel and gives its lesson (Rule 9).

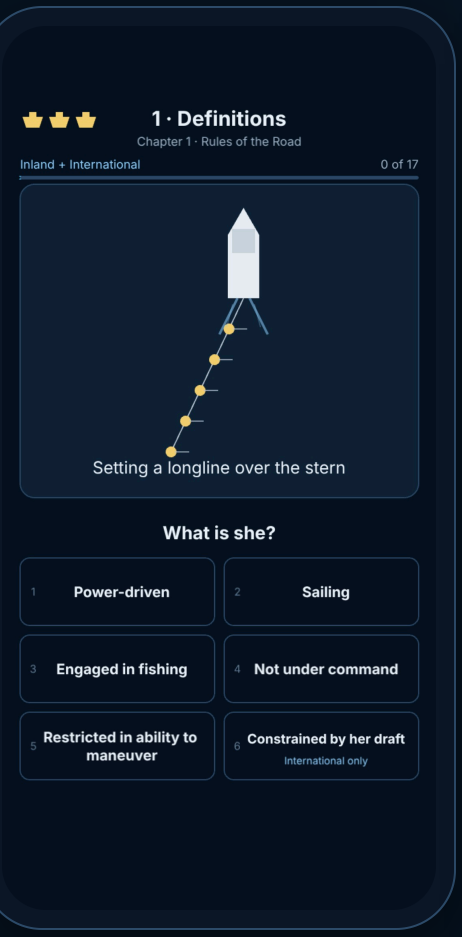
In the level

- Five gates in order, three of them rounding anchored boats of different sizes.
- A powerboat crosses from starboard, a ferry meets you head-on and turns to its own starboard, a kayak crosses near the slip.
- Any contact with traffic ends the run, even at rest.
- The slip counts only after the last gate. Par 150 s.
- Golden replays prove a clean docking and a turn to port into the ferry.

 Real build

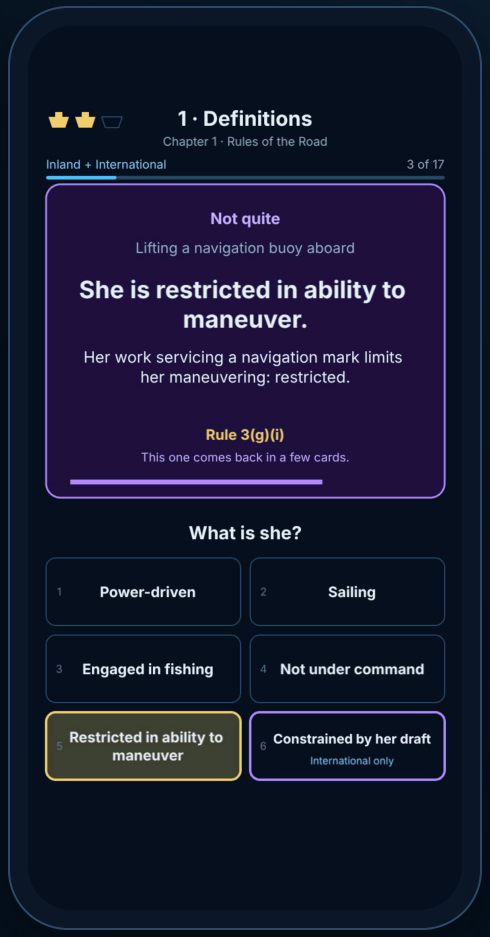
The first Rules of the Road level, built

Branch head-dev/ch1-definitions, issue #47, at /rules. Upright, one thumb.



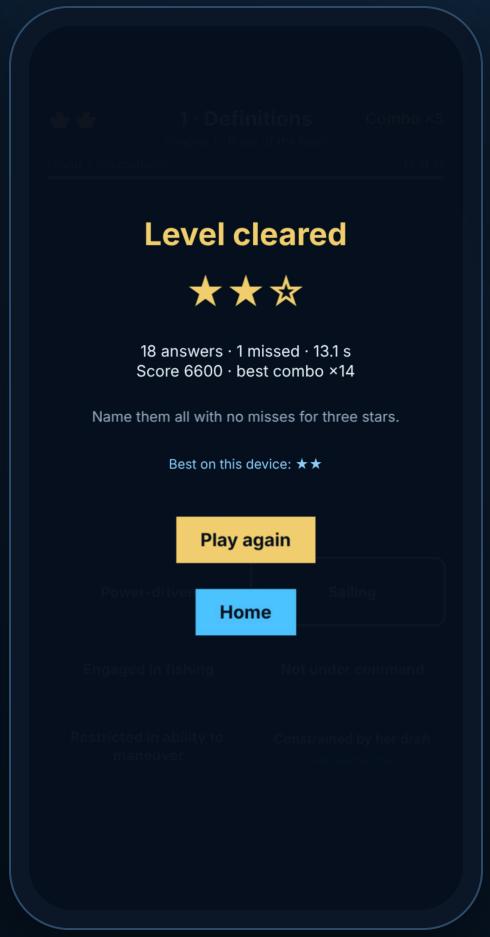
A card

vessel drawn top-down with its cue, and six answers.



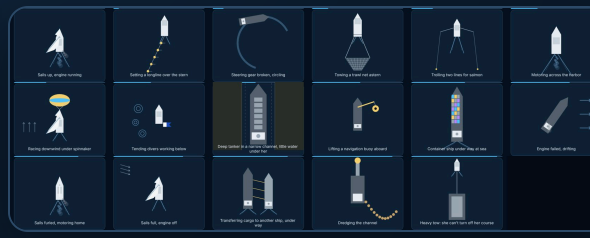
A miss

A heart lost; the replay shows the answer, reason and Rule 3(f).



Cleared

Two stars for one miss; Play again or Home.



All 17 vessels

Including the trick cards: motor-sailing and trolling are power-driven.

Where the prototypes stand

Two levels are built with tests and waiting on pull requests. The rest are proposals.

Status

Prototype	State	Ticket
Chapter 1 mockups, decks, Harbor Run course	Mockup, proposal	-
Night Harbor Run	Playable prototype, approved, not in the repo	#45
Level 3 harbor course	Real build on its branch, tests green, no pull request	#46
Chapter 1, level 1 Definitions	Real build on its branch, tests green, no pull request	#47

Next steps

1. Playtest Definitions and level 3 with about five license candidates on their own phones.
2. Have uscg-regs check every card, lesson and citation against 33 CFR 83 and COLREGS.
3. Fix what both turn up, then build Chapter 1, level 2 (Lights) from the lights deck.
4. Answer the four questions on #45 before Night Harbor Run moves into the repo.